P13: Investigating the behavior of internal tides at Cape Hatteras using inverted echo sounders

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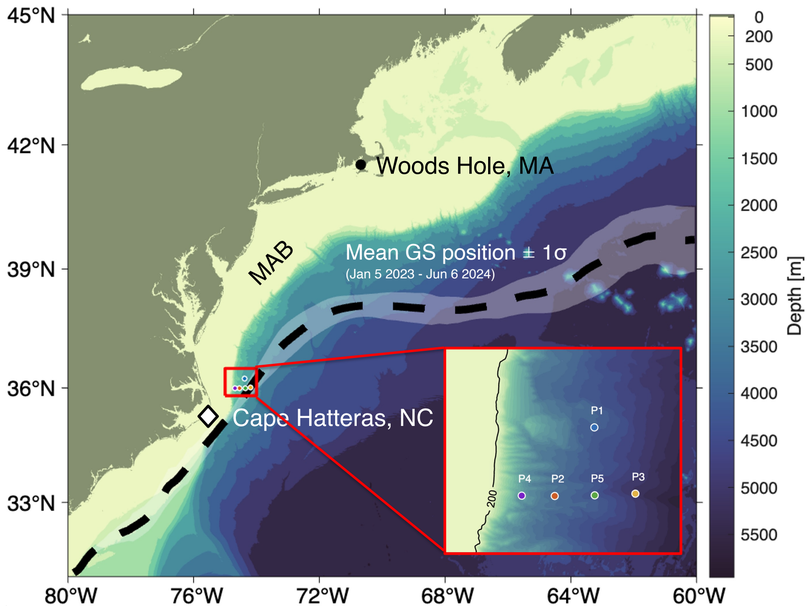


Figure 1. MAB-SWOT Region and the location of the MAB-SWOT CPIES array (red box and inset).

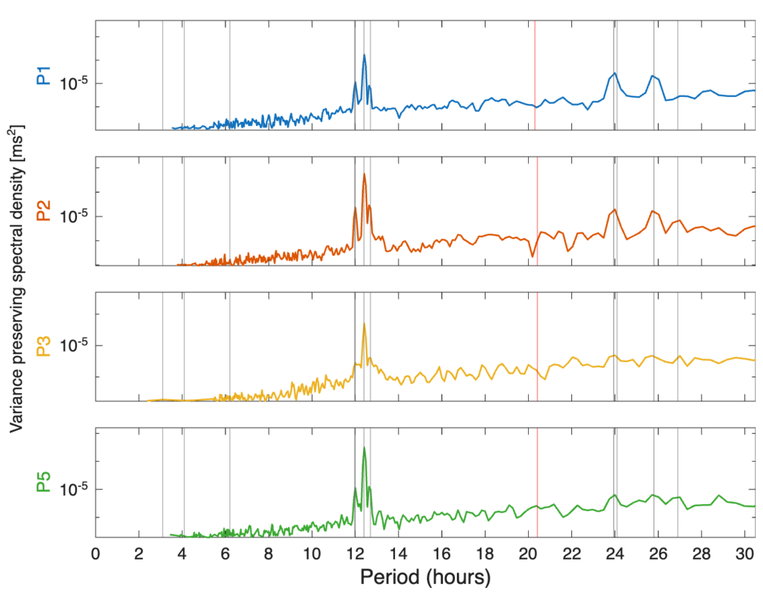
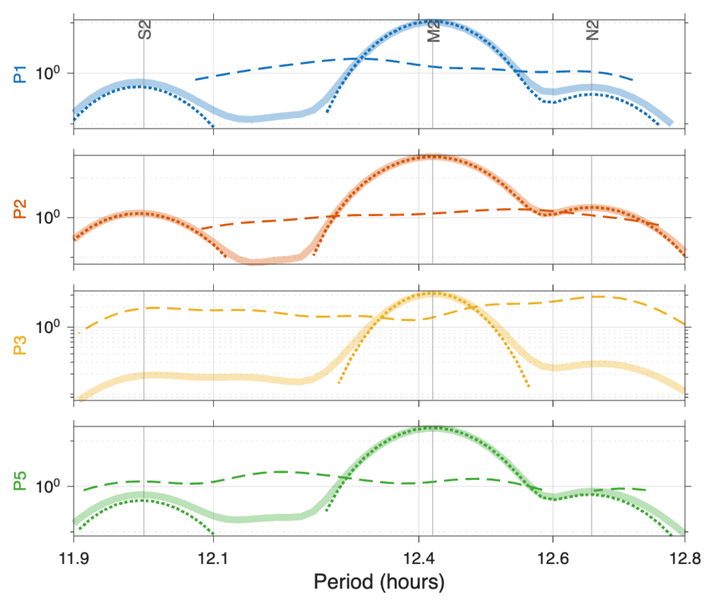


Figure 2. Full spectra of ITs (τ_{res}) at each CPIES; Figure 3. Semidiurnal spectra of full, coherent, and incoherent ITs



1 The MAB–SWOT region

The Mid-Atlantic Bight (MAB)

- a region on the US East Coast stretching from Cape Hatteras, NC to Cape Cod, MA
- Largely owing to the Gulf Stream, it is highly energetic, with a constantly evolving mesoscale and submesoscale environment
- This modifies the internal wave field

Surface Water and Ocean Topography (SWOT)

- SWOT (launched 2022) maps SSH in 2-D
- Resolves submesoscale processes as low as 15 km

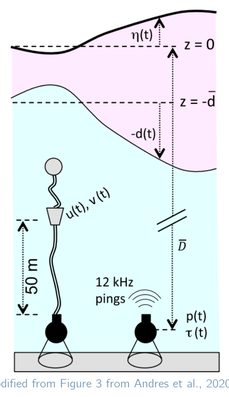
The SWOT cal/val period

From March - July 2023, SWOT performed 2 daily swaths over the MAB-SWOT CPIES array (Fig. 1)

https://www.swot-adac.org/campaigns/mab-swot/

2 Inverted echo sounders

- 6 current and pressure sensor equipped inverted echo sounders (CPIESs) deployed January 2023 to June 2024 in the MAB-SWOT region
- 4 CPIES (Fig. 1) recorded hourly
  1. **τ: round-trip acoustic travel time**
     - sensitive to the mode-1 internal tide
     - $\tau = 2\int \frac{dz}{c(z)}$ where c(z) is the sound speed profile
  2. **prs_{bottom}: bottom pressure**
     - sensitive to barotropic tide



Modified from Figure 3 from Andres et al., 2020

3 Internal tides (ITs)

- Form when the barotropic tide interacts with sloping topography
- Coherent ITs stay phase-locked to the forcing; incoherent ITs wander
- ITs are strongly incoherent in the MAB-SWOT region
  - Modifications from currents and fromts (e.g., Gulf Stream, Hatteras Front, shelfbreak front) (Kelly & Lermusiaux 2016; Savidge et al. 2007)
  - Mode-1 IT decorrelation is ~5 days; >55% of sea level variance is due to incoherent ITs (Lahaye et al., 2024)

ITs obscure SWOT measurements

Corrections are made by the High Resolution Internal Tides (HRET) model that predicts coherent tides

4 Synthesis: SWOT + CPIES

Identify incoherent ITs with the CPIES during the cal/val period to compare with HRET corrections and obtain a corrected SWOT product

5 IT behavior from the CPIES

- ITs (τ_{res}) are mostly coherent at each CPIES, indicating local generation
- M2 is the dominant tidal period
- Incoherent ITs are spread between K2 (11.97 hours) and N2 (12.66 hours)

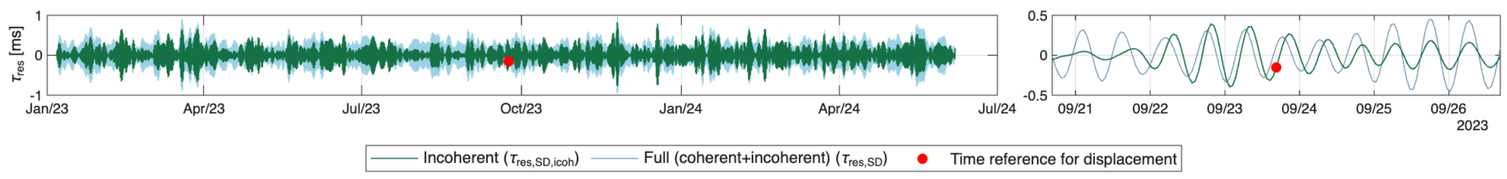


Figure 4. Full (left) and zoomed in (right) timeseries of the full and incoherent τ_{res} at P5.

6 IT impact on the water column & SSH

How to get from $\tau_{res}(t)$ to the estimated $SSH_{IT}(t)$

Calculate the stratification $N^2(z)$ of a background density profile $\rho(z)$

↓

Solve the following equation to get the IT mode-1 vertical structure $W_1(z)$: $\frac{d^2W_1(z)}{dz^2} + \frac{N^2(z)}{c_1^2}W_1(z) = 0$

↓

Calculate the linear transfer function T that converts τ [seconds] to pycno-cline displacement

- Displace ρ(z) by some amplitude η x $W_1(z)$
- Compute τ(η) for the $\rho_{displaced}(z)$ profile
- Repeat for another η

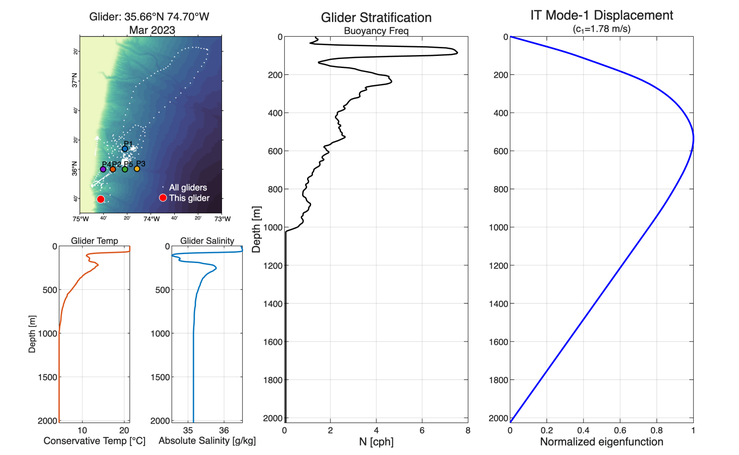
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Apply T to $\tau_{res}(t)$ to get the perturbed density profile ρ'(z,t)

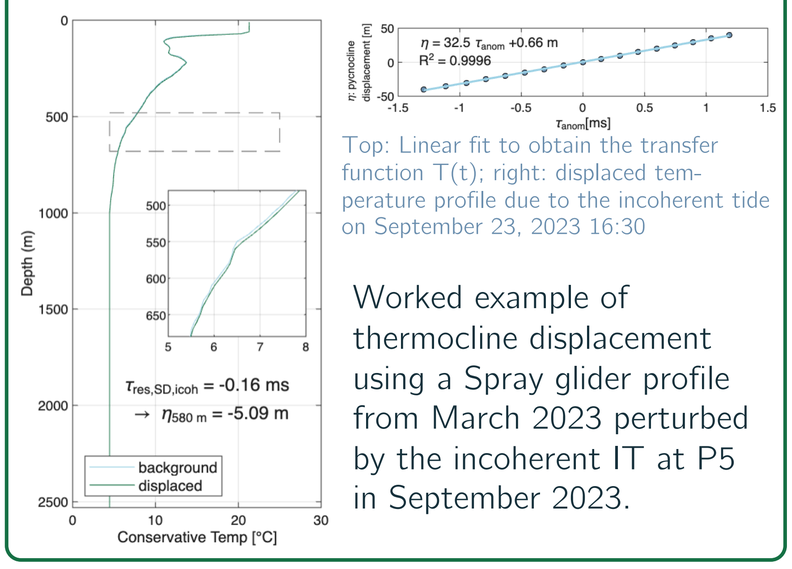
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Calculate the SSH_{IT} (steric SSH of ρ'(z,t)) and compare with SWOT

7 Example with a glider profile



Background profiles from a Spray Glider



Top: Linear fit to obtain the transfer function T(t); right: displaced temperature profile due to the incoherent tide on September 23, 2023 16:30

Worked example of thermocline displacement using a Spray glider profile from March 2023 perturbed by the incoherent IT at P5 in September 2023.

8 Ongoing & future work

- Map $\tau(t) \rightarrow \rho(z) \rightarrow T(t) \rightarrow \rho'(z,t) \rightarrow SSH_{IT}(t)$ at each CPIES during the cal/val period
- Compare $SSH_{IT}(t)$ with HRET
- Quantify how much of the SWOT SSH variance is explained by the incoherent IT
- Diagnose if a flat bottom assumption suffices at each CPIES

References & acknowledgements

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